

INDIAN INSTITUTE OF TECHNOLOGY ROPAR

CH512: Machine Learning Applications to Process Engineering

AY2025-2026

Course Project Part 2: Nonlinear function approximation techniques

(Try to implement as much as you CAN)

1. **Parametric function approximation techniques**: Develop an **optimal nonlinear predictive model** or the **soft sensor/virtual sensor** for the chosen target variable based on neural networks by considering the following steps:
 - Input Variable / Feature selection: **(results that you have obtained in Project-1 to be used in continuation)**
 - i. Forward selection method or correlation method or subset selection method/k-fold validation method.
(or)
 - ii. PCA based feature selection.
 - Model building:
 - i. Build the following neural network models
 - **Feed forward neural network model (Conventional NN model)**. Here, compare the performance of shallow neural network (single hidden layer), with deep neural networks (multiple hidden layers).
 - Hyperparameter tuning: Check the performance of the conventional ANN for (a) different learning rates (b) different values of regularization parameters (c) different hidden layers (horizontal and vertical).
 - **Autoregressive based Deep NN Model**: Develop an autoregressive based Deep NN model. Here, *in addition to the input variables as features, include onetime delayed, and two-time delayed output variables as additional input variables (or feature) into the model.*

- **LSTM models:** Develop an LSTM model using the inputs selected from any of the feature selection techniques mentioned before.
 - Validation of the developed NN models on the test data sets. From total data, consider 70% as training data and 30% as validation data sets.
2. **Non-Parametric function approximation techniques:** Develop an **optimal just-in-time learning based predictive model** or the **soft sensor/virtual sensor** for the chosen target variable based on neural networks by considering the following steps:
- Input Variable / Feature selection: **(results that you have obtained in Project-1 to be used in continuation)**
 - i. Forward selection method or correlation method or subset selection method/k-fold validation method.
 - Model building:
 - i. From total input and output data, consider 70% as training data and 30% as validation data sets.
 - ii. Create a database which includes the training data, with inputs and its corresponding values of the target variables.
 - iii. For the validation data, for each validation data point (i.e., for every validation of input variable), use k-NN search algorithm to select k nearest neighboring input data points ($\mathbf{u}$) from the training data base. Select corresponding target variables ($\mathbf{y}$).
 - iv. Based on the k input and output data points ($\mathbf{u}, \mathbf{y}$), build a local just-in-time based neural network model (1 hidden layer NN model) using k-NN search algorithm.
 - v. Using the developed local model carry out the predictions at the current validation input data point. For the next validation input data

point, discard previous model and follow steps (ii, iii). Repeat the procedure for all the data points.

3. Presentation of the results:

- Compare the predictive performance of all the developed models (Project Part1 and Part 2) using the following metrics: Sum of squared errors (SSE), R^2 .
- Residuals: Plot residuals of different models (i.e. $y_{\text{test}} - y_{\text{model_predicted}}$) vs No. of samples / No. of test data points chosen. Find auto-correlation coefficient (i.e., correlation of residuals between successive time instants), and plot auto-correlation coefficient w.r.t. residuals. What do you infer from this analysis?
- True and Model Prediction comparison plots, w.r.t., time for all the models developed.
- Compare the predictive performance of the developed models (i.e., $y_{\text{predicted}}$ vs y_{test} data) through scatter plots.